

Road Pavement Defect Detection Using YOLOv8

1. Problem and Objective

Road pavement defects such as potholes, cracks, and uneven manholes can affect road safety, ride quality, and maintenance requirements. Traditional road condition assessment relies heavily on visual inspection, which can become time-consuming when applied across large road networks.

This project explores YOLOv8 object detection as a computer vision approach for preliminary road-condition screening. The objective is to develop and evaluate a prototype capable of detecting three visible defect classes: pothole, road_crack, and uneven_manhole.

The model is intended to assist inspection and maintenance personnel by identifying locations that may require further investigation rather than replacing professional road inspection.

2. Dataset Preparation

The dataset was assembled from selected images obtained from two publicly available Kaggle road-damage datasets, together with additional road photographs taken for the project. Images were curated and annotated in Roboflow using three classes: pothole, road_crack, and uneven_manhole. The uneven_manhole class also includes visually identifiable open manholes.

Dataset Version 1 contained 252 source images. Images were auto-oriented and resized to 640 x 640 pixels. Training augmentation included horizontal flipping, brightness adjustment (+/-15%), and blur (up to 1.9 px). After augmentation, the exported dataset contained 606 training images and 50 validation images, giving 656 images in total.

SAM was briefly explored as an annotation-assistance tool. It was useful for some clearly visible defects but thin and irregular road cracks often required manual correction. Five additional project photographs were kept completely outside training and validation and reserved for unseen image inference.

3. Model Development and Reproducibility

The model was developed using YOLOv8 Nano (yolov8n.pt) in Google Colab. Training was completed for 30 epochs using a batch size of 16 and an image size of 640 x 640 pixels, with an NVIDIA Tesla T4 GPU used during the final reproducibility test.

The trained best.pt weights were retained for inference. Reproducibility was supported through a public GitHub repository containing separate training and inference notebooks, together with a frozen GitHub Release containing the dataset and trained weights. Both notebooks were successfully tested in Google Colab without relying on local computer files.

Dataset -> YOLOv8 Training -> Validation -> Error Analysis -> Unseen-Image Inference

4. Model Results

The YOLOv8 model achieved the following overall validation results:

Metric	Result
Precision	0.511
Recall	0.434
mAP@50	0.438
mAP@50-95	0.207

Among the three classes, uneven_manhole showed the strongest performance, while road_crack was the most challenging class. The results demonstrate that the prototype can detect visible pavement defects, but further improvement is required for more consistent performance.

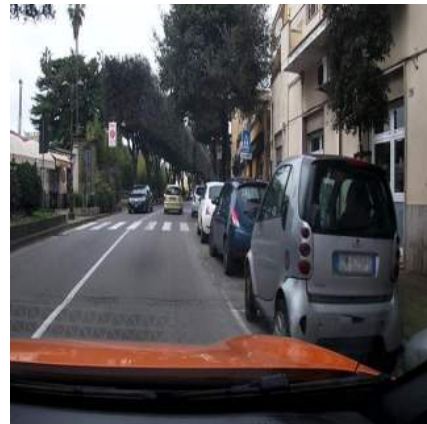
5. Error Analysis and Limitations

The main errors observed were missed road cracks, missed small or distant potholes, and occasional class confusion. Road cracks were particularly challenging because of their thin and irregular appearance.

Performance may also be affected by shadows, road markings, repaired surfaces, viewing distance and pavement texture. The current model was mainly trained using daytime road images and has not been sufficiently tested under night-time, rain, wet-road or poor visibility conditions. The model should therefore be treated as a preliminary screening tool, with human verification required before maintenance or safety related decisions.



Example success: pothole



Example failure: missed road crack

6. Conclusion and Future Improvement

The project demonstrates the potential of YOLOv8 for preliminary road pavement defect screening. The prototype detected examples from all three classes, while evaluation also identified limitations requiring further improvement.

Future development should focus on increasing road crack diversity, adding more challenging environmental conditions such as night and rain, and improving annotation consistency. For practical field use, the approach could later be extended to vehicle-mounted cameras with GPS to support defect location and maintenance planning.

This model is an assistive tool for preliminary screening only. It produces False Negatives and must not be used as the sole verifier for life-safety decisions.